PRACTICAL – 10

Asynchronous Processing with Event-Driven Architecture using Node.js EventEmitter.

Objective

The objective of this practical is to:

1. Implement asynchronous background processing using Node.js's built-in EventEmitter.
2. Create a custom event for newly created tasks.
3. Decouple notification processing from the main API request.
4. Demonstrate that the API response can be sent before the background notification handler completes.
5. Understand the practical working of Node.js event-driven architecture. 6. Implement additional task-deleted and error events

Step 1 — Create events.js



2 Create listeners.js

listeners.js U

backend > listeners.js > ...

```
const eventManager = require("./events");

// ==========================================
// EVENT CREATED LISTENER
// ==========================================

eventManager.on("event-created", (event) => {

  console.log(
    `[Notification] Event "${event.title}" created at ${new Date().toISOString()}`
  );

  // Simulate slow background notification processing
  setTimeout(() => {

    console.log(
      `[Notification Completed] Event "${event.title}" processed at ${new Date().toISOString()}`
    );

    console.log(
      `[Notification] Location: ${event.location}`
    );

  }, 2000);

});
```

```
// ==========================================
// EVENT DELETED LISTENER
// ==========================================

eventManager.on("event-deleted", (event) => {

  console.log(
    `[Deleted Notification] Event "${event.title}" was deleted at ${new Date().toISOString()}`
  );

});

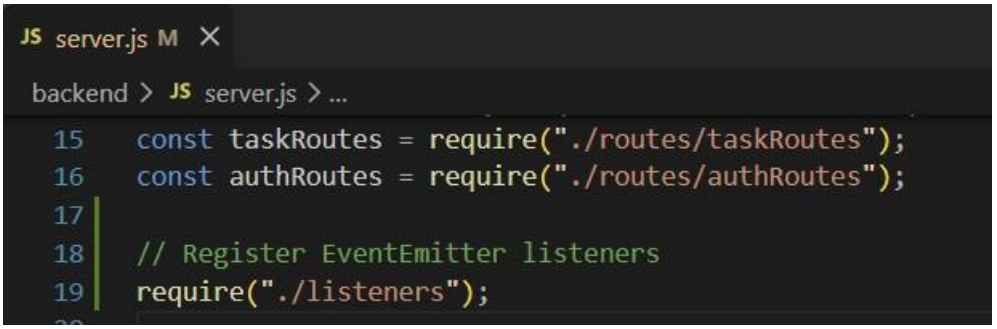

// ==========================================
// ERROR LISTENER
// ==========================================

eventManager.on("error", (error) => {

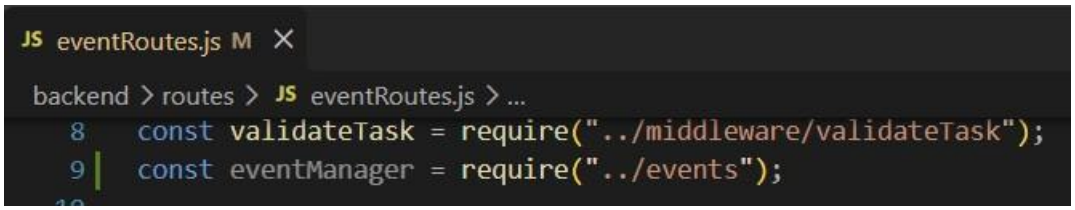
  console.error(
    `[Event Error] ${new Date().toISOString()} - ${error.message}`
  );

});
```

3 Register the listeners in server.js



Step 4 — Modify your existing eventRoutes.js



Step 5 — Modify POST /api/events

```
// POST create new event (Protected with authMiddleware & validateTask)
router.post("/", authMiddleware, validateTask, async (req, res, next) => {
  try {

    // Save event to MongoDB
    const event = await Event.create(req.body);

    console.log(
      `[API] Event created at ${new Date().toISOString()}`
    );

    // Send API response immediately
    res.status(201).json({
      success: true,
      message: "Event created successfully",
      data: event,
    });

    // Emit event after sending response
    eventManager.emit("event-created", event);

  } catch (error) {
    next(error);
  }
});
```

6 Modify DELETE /api/events/:id

```
// DELETE event (Protected with authMiddleware)
router.delete("/:id", authMiddleware, async (req, res, next) => {
  try {

    const event = await Event.findByIdAndDelete(req.params.id);

    if (!event) {
      return res.status(404).json({
        success: false,
        message: "Event not found",
      });
    }

    console.log(
      `[API] Event deleted at ${new Date().toISOString()}`
    );

    // Send response immediately
    res.status(200).json({
      success: true,
      message: "Event deleted successfully",
    });

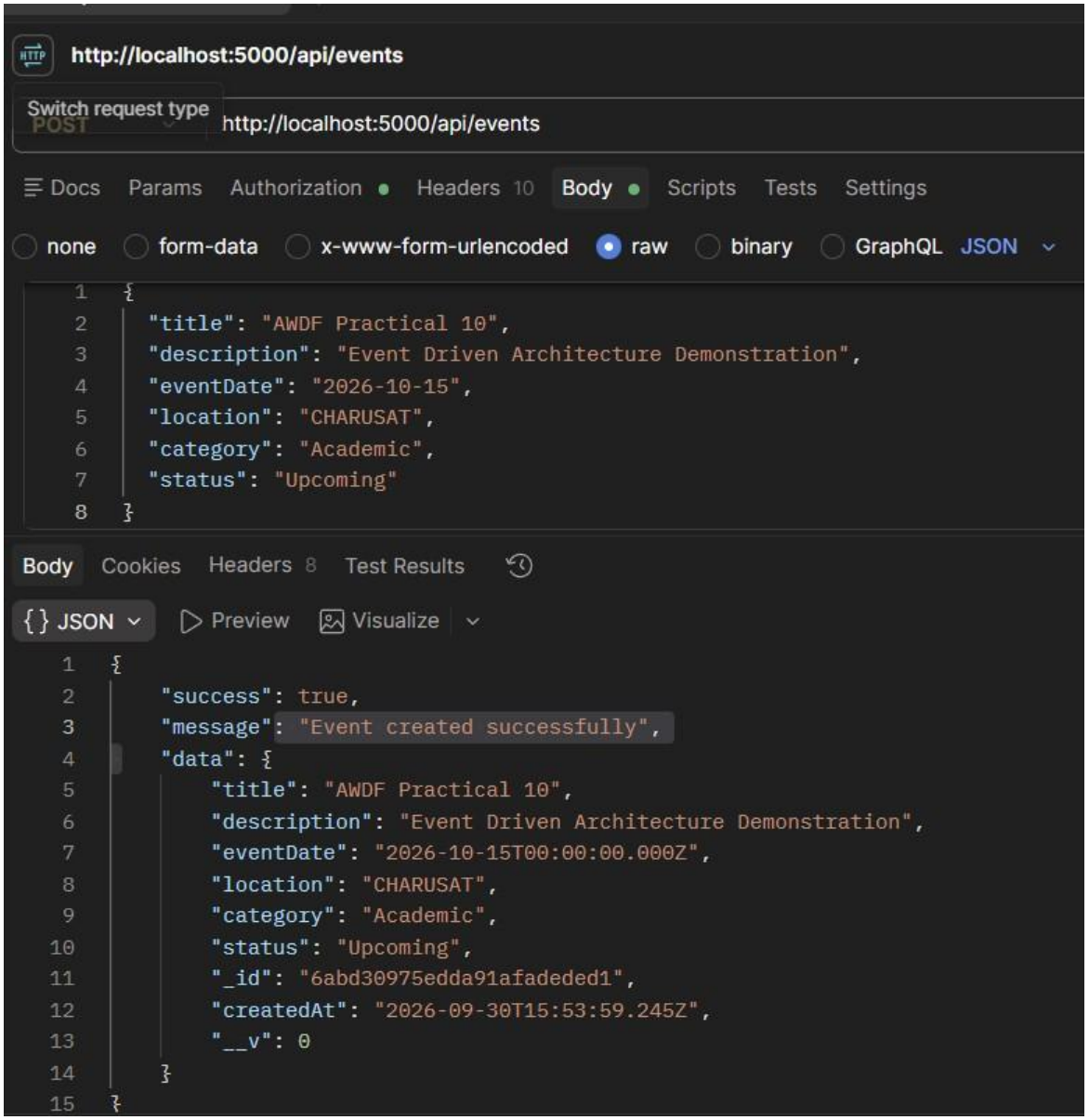
    // Emit delete event
    eventManager.emit("event-deleted", event);

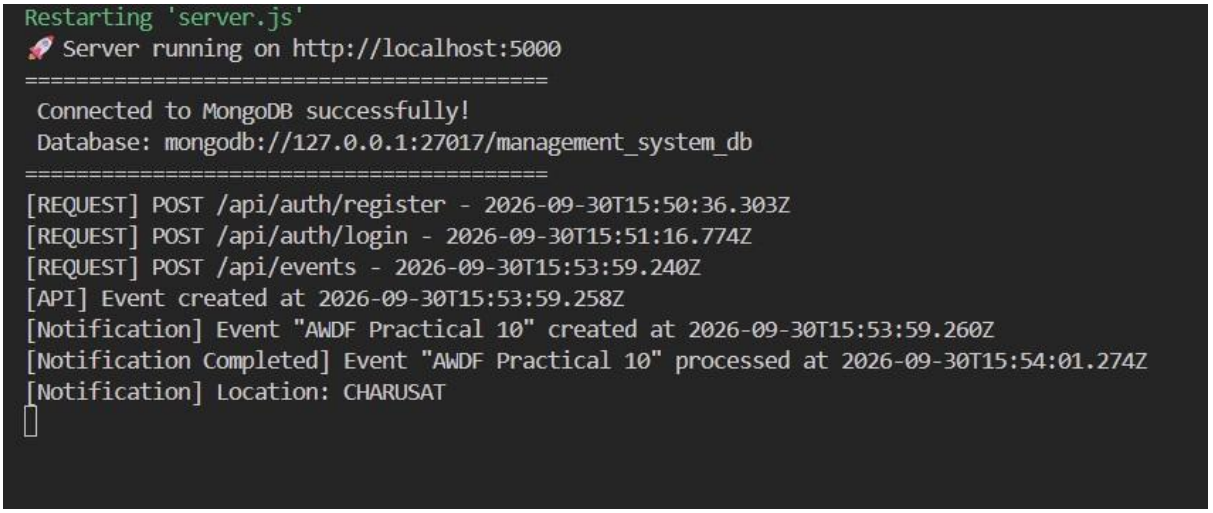
  } catch (error) {
    next(error);
  }
});
```

Step 7 — Start the backend

```
Restarting 'server.js'
🚀 Server running on http://localhost:5000
=======================================
 Connected to MongoDB successfully!
 Database: mongodb://127.0.0.1:27017/management_system_db
=======================================
```

8 Test event creation


http://localhost:5000/api/events
Switch request type
POST
http://localhost:5000/api/events
Docs Params Authorization Headers 10 Body Scripts Tests Settings
none form-data x-www-form-urlencoded raw binary GraphQL JSON
{
"title": "AWDF Practical 10",
"description": "Event Driven Architecture Demonstration",
"eventDate": "2026-10-15",
"location": "CHARUSAT",
"category": "Academic",
"status": "Upcoming"
}
Body Cookies Headers 8 Test Results
{ } JSON Preview Visualize
{
"success": true,
"message": "Event created successfully",
"data": {
"title": "AWDF Practical 10",
"description": "Event Driven Architecture Demonstration",
"eventDate": "2026-10-15T00:00:00.000Z",
"location": "CHARUSAT",
"category": "Academic",
"status": "Upcoming",
"_id": "6abd30975edda91afadeded1",
"createdAt": "2026-09-30T15:53:59.245Z",
"__v": 0
}
}



Github: https://github.com/sahilmak1602/WDF_24IT045_SAHIL.git